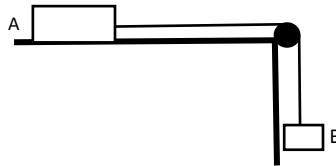


Dynamics: Exercises Part II

Exercise 1

A block A of metal of mass 2 kg on a horizontal table is attached another block B of mass of 0.45 kg by a light string passing over a frictionless pulley at the edge of the table. By allowing the 0.45 kg mass to fall the block A is subjected to an horizontal force. The coefficient of sliding friction between the block and table is 0.2. Calculate (a) the initial acceleration, (b) the tension in the string, (c) the distance the block would continue to move if, after 2 s of motion, the string should break.

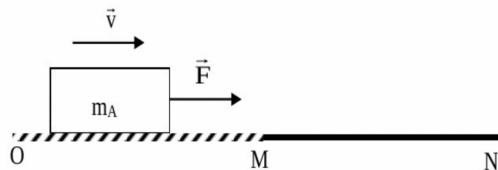


Exercise 2

A point-like object A, of mass $m_A = 2 \text{ Kg}$ moves on a flat surface having two parts. A rough one from O to M and a smooth one from M to N. The contact between (A) and the rough surface is characterized by static and kinetic friction coefficients $\mu_s = 0.6$ and $\mu_d = 0.4$, respectively. What is the minimum force to start the motion.?

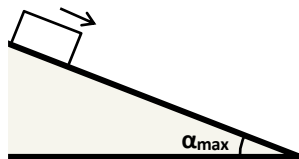
The object has a velocity $\vec{V}$, we exert a constant force $\vec{F} = 14(N)\vec{i}$ parallel to $\vec{V}$.

- 1- Sketch forces diagram exerted on (A) when the mobile moves from O to M.
- 2- Find the acceleration. What is the motion type and its time equation?
- 3- If we consider that at O we have $v = 2 \text{ m/s}$, Find the mobile velocity at M if $OM = 5 \text{ m}$?
- 4- At M a collision holds between A and another mobile B of mass $m_B = 8 \text{ kg}$, at rest. After the choc the two objects remain stick. Find velocity vector of the system (A+B) after the choc. ($g = 10 \text{ m/s}^2$)



Exercise 3 (from Final exam 2024)

A block of mass $m = 1 \text{ kg}$ is placed at the top of inclined plane with a varying inclination α and a rough surface characterized by a static and kinetic coefficients $\mu_s = 0.4$ and $\mu_d = 0.2$ respectively.



- 1- Draw the forces acting on the body then show that the maximum value of α that allow to keep the body still at rest (before a motion start) fulfill the condition:

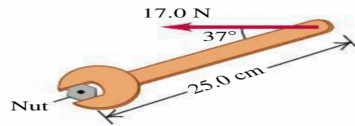
$$\tan(\alpha_{max}) = \mu_s$$

- 2- The body is pushed by with an initial velocity $v = 2 \text{ m/s}$ parallel to the inclined plane, what is the magnitude of its acceleration and the distance travelled before it stops.
- 3- if one wants to move the body with constant velocity, what it is the magnitude of the force that should be applied on it.

Exercise 4

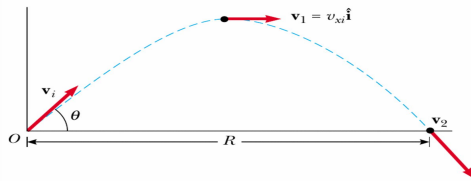
A machinist is using a wrench to loosen a nut. The wrench is 25 cm long, and he exerts a 17 N force at the end of the handle at 37° with the handle.

1. What torque does the machinist exert about the center of the nut?
2. What is the maximum torque he could exert with this force, and how should the force be oriented?



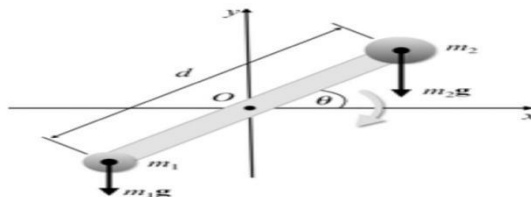
Exercise 5

5- A particle of mass $m=1\text{Kg}$ is shot with an initial velocity $v_i=5\text{ m/s}$ making an angle $\theta=30^\circ$ with the horizontal as shown in Figure. The particle moves in the gravitational field of the earth. Find the angular momentum of the particle about the origin when the particle is (a) at the origin, (b) at the highest point of its trajectory, and (c) just before it hits the ground. (d) What torque causes its angular momentum to change? What is the expression of the tangential F_T and the normal F_N components of the force applied on the particle at any time? ($g=10\text{ m/s}^2$)



Exercise 6

A rigid rod of mass M and length d is pivoted without friction about its center, as shown in the Figure. Two particles of mass m_1 and m_2 , respectively, are connected to its ends. The combination rotates in a vertical plane with an angular speed ω . Give an expression for (a) the magnitude of the angular momentum of the system, and (b) the magnitude of the angular acceleration of the system when the rod makes an angle θ with the horizontal direction



Exercise 7

A uniform solid disk of mass 3 kg and radius 0.2 m rotates about a fixed axis perpendicular to its face. If the angular frequency of rotation is 6.00 rad/s, calculate the angular momentum of the disk when the axis of rotation (a) passes through its center of mass and (b) passes through a point midway between the center and the rim (use Huygens–Steiner theorem).